



BASE24[®]

Dual Site Processing Guide

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Preface

This manual reviews the BASE24 Release 6.0 Version 4 Dual Site Support enhancements and makes recommendations relating to a Dual Site implementation utilizing GoldenGate for data replication.

Audience

The configuration and maintenance of a Dual Site environment is a complex activity that will involve a variety of technical personnel. This manual is written with the BASE24 and HP NonStop systems administrators as the intended audience.

Additional Documentation

This manual contains references to the following documents.

BASE24 Release 6.0 Version 4 Implementation Guide

Publication Number: BA-AE010-04

This document provides details of the enhancements made in Release 6.0 Version 4 to provide support for BASE24 Dual Site operation.

GoldenGate Dual Site Processing Guide

This document provides details of the GoldenGate products and configuration required to complement the BASE24 Dual Site Support features.

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Section 1

Introduction

This Guide reviews the Dual Site Support enhancements released as part of BASE24 Release 6.0 Version 4, discusses some of the issues related to the running of BASE24 in a Dual Site production environment and describes one possible implementation. This Guide should not be viewed as a step-by-step guide to implementing a Dual Site environment. Such a project will continue to be lengthy and complex, with many possible solutions. This guide intends to assist someone planning such a project to see how the new BASE24 features could be used to help design and build such a solution. Consultancy from ACI and / or GoldenGate may be required to help design and implement a Dual Site environment.

It should be noted that the enhancements made available in Release 6.0 Version 4 do not enable or facilitate Dual Site operations in and of themselves. For instance, the enhancements do not provide for the replication of transactions or data between two sites. It is assumed that a third party product will undertake this. The enhancements are intended to provide support for processing options that will allow and simplify the configuration and operation of a Dual Site environment.

A BASE24 Dual Site production environment is where BASE24 is running on two separate HP NonStop machines, with production transactions being processed on both machines. Often the two systems will have duplicate HP NonStop and BASE24 configurations and either site will be capable of processing the full transaction load, but neither of these features are required. It may be that the two systems have minor configuration differences in terms of communications connections or disc types and distribution, but it may also be that the two systems have very different configurations including having different types and or numbers of CPUs. In either case Dual Site operation is still possible. It is also possible that either one or both of the systems would be unable to process the full transaction load at peak times. In such cases it may be that the full load could be processed at non-peak times. In such scenarios it would be necessary to have some criteria to control the number and sequence of terminals and connections that are started in the event of a system failure.

The bi-directional replication of configuration, customer and transaction data between the two sites is fundamental to the functioning of a Dual Site system. Configuration and security files may or may not be replicated depending upon whether the two systems are exact duplicates. The major files that require replication are the customer authorization files (CAF, PBF etc.) and the transaction log files (TLF and PTLF).

Bi-directional replication (to and from the same files) is not supported by all of the available data replication tools. Additionally, the accurate replication of Authorization data requires that changes to data rather than the data itself is replicated. This is to avoid the situation where changes are made concurrently on the two systems (possibly by joint account holders making withdrawals at the same time) and the new data values are both replicated to the alternate system resulting in both systems having incorrect data. GoldenGate supports bi-directional replication and the replication of data changes rather than final values (where appropriate) and was therefore selected as the data replication utility upon which the BASE24 Dual Site Support design was based.

The functionality provided by GoldenGate, and assumed by the BASE24 Release 6.0 Version 4 enhancements, is referred to in this document, but the GoldenGate modules and configuration required to achieve this are not discussed. For details on the use of GoldenGate to support a Dual Site environment, please refer to the GoldenGate Dual Site Processing Guide.

Section 2

Release 6.0 Version 4 Dual Site Support

This section reviews the code enhancements made as part of BASE24 Release 6.0 Version 4 to support Dual Site operations.

Settlement

ATM and POS Settlement processes have been enhanced to support process specific TLF / PTLF assigns, broadcast capability in notify lists and to recognize and optionally process local only, remote only or all terminals and transactions.

Please see the BASE24 Release 6.0 Version 4 Implementation Guide for details of the LCONF params and assigns required by the Settlement processes for Dual Site Support.

Extract

The Extract process has been enhanced to support process specific TLF / PTLF assigns and to recognize and optionally process local only, remote only or all terminals and transactions.

Please see the BASE24 Release 6.0 Version 4 Implementation Guide for details of the LCONF params and assigns required by the Extract process for Dual Site Support.

Refresh

The Refresh process has been enhanced to support process specific TLF / PTLF assigns, broadcast capability in notify lists and to recognize and optionally process local only, remote only or all transactions during impacting.

Please see the BASE24 Release 6.0 Version 4 Implementation Guide for details of the LCONF params and assigns required by the Refresh process for Dual Site Support.

TLF / PTLF Perusal

ATM and POS Transaction Log File Perusal Server processes have been enhanced to support process specific TLF / PTLF assigns and to recognize and optionally identify remote transactions.

Please see the BASE24 Release 6.0 Version 4 Implementation Guide for details of the LCONF params and assigns required by the TLF / PTLF Server processes for Dual Site Support.

TLF Perusal Screen

TLF Perusal screen 2 contains a flag that identifies remote transactions, as shown.

BASE24-ATM	TRAN LOG LIST	SYSF	BICI	03/10/09	16:58	02 OF 02
CARD:		FIID:	BICI	MEMBER NUM:	***	
TERMINAL:	TABCDEF	LNET:	BICI	DATE (YYMMDD):	031006	
TRAN CODE:	100100	W/D FROM CHK	DEPOSITORY:			
TRAN DATE:	06/10		TRAN TIME:	09:32:57	TIME OFST:	0
FROM ACCT:			AMOUNT 1:	9.00		
TO ACCT:			AMOUNT 2:	0.00	GBP	
			AMOUNT 3:	0.00		
SEQ NUM:	200002110004		DEP BAL CR:	0.00		
TERMINAL:	ABCDEF					
PAN:	5603773773120082	TERM LNET:	BICI			
RESP CODE:	0001 APPROVED-NO BALANCES	BRANCH:				
AUTH ID RESP:		REGION:				
RVSL REASON:	00	ERR FLAG:	0			
RCPT REQUESTED:						
ICHG PROC CDE:						
TXN SUBTYPE:		REMOTE TXN				
NO MORE DETAIL INFORMATION IS AVAILABLE FOR THIS TRANSACTION.						
***** BASE24 *****						
FILE DESTINATION: NEW LOGICAL NETWORK ID:						
F5-GOTO SUMMARY F6-NXT REC F7-PRV REC F9-NXT PAGE F11-PRV PAGE F12-HELP						

PTLF Perusal Screen

PTLF Perusal screen 4 contains a flag that identifies remote transactions, as shown.

BASE24-POS	PTLF DETAIL	SYSF	BNK1	03/10/09	16:55	04 OF 04
PERUSAL TYPE/REC FRMT: D DATE(YMMDD): 031009 FIID: BNK1 LNET: SYSF CARD: 5603773773120091 MEMBER NUM: 000 TERMINAL: TERMSPDH1 RETAILER: MERCHSPDH CLERK:						
PAN: 5603773773120091 MEMBER NUM: 000 EXP DATE: 0312 CARD TYPE: M ISS CODE: 00 BATCH NUM: 001 TERM: TERMSPDH1 SIC CODE: 0000 SHIFT NUM: 001 TRAN DATE: 08/10 TIME OFST: 0 TRAN TIME: 16:59:10 TRAN: 101310 PURCHASE ERR FLAG: PIN PRESENT: N DRAFT CAP: 1 (AUTH & CAPTURE) APPRV CODE: 162941 B INVOICE: 0000000000 ACCT: OVERRIDE FLAG: 0 ORIG INVOICE: 0000000000 AUTHORIZER: AUTHH AUTH IND: P SEQ NUM: 000001001006 RESP CODE: 3001 (APPROVED-NO BALANCES) AUTH SEQ NUM: 000000000000 AMT 1: 10.00 INDUSTRY FRMT: AMT 2: GBP CONV FLG: ICHG PROC CDE: ALT MRCHNT: TXN SUBTYPE:						
						REMOTE TXN
***** BASE24 *****						
FILE DESTINATION:			NEW LOGICAL NETWORK ID:			
F5-GOTO SUMMARY	F6-NXT REC	F7-PRV REC	F9-NXT PAGE	F11-PRV PAGE	F12-HELP	

ATD / PTD Management

ATM and POS Terminal Definition File Server processes have been enhanced to display and maintain a flag identifying terminals as local or remote.

Please see the BASE24 Release 6.0 Version 4 Implementation Guide for details of the LCONF params and assigns required by the ATD/PTD File Server processes for Dual Site Support.

ATD Screen 1

ATD screen 1 contains the Dual Site Indicator field, as shown.

BASE24-ATM		ATM TERMINAL DATA		SYSF BNK1		03/10/16 11:06		01 OF 21	
TERM ID: P1A^NCR5022		FIID: BNK1		REGION:		BRANCH:		LN: SYSF	
DEVICE TYPE: 22 (NCR 5XXX)		PHONE:		LOG ROUTING CODE:		0			
				DUAL SITE INDICATOR:		R (REMOTE)			
OWNER NAME:		LOCATION:		STATE:					
CITY:									
COUNTRY:		POSTAL CODE:							
DH PROCESS: P1B^N50		AUTH PROCESS: P1A^AUTH1							
INST ID NUM: 000000000000		ALT ROUTING GROUP: 000000000000							
		NCD ROUTING GROUP: 000000000000							
SHARING INFORMATION									
COUNTY CODE: 0		STATE CODE: 0		COUNTRY CODE: 826					
GROUPS:				SURCHARGE PROFILE:					
POSTING DATE:		TIME OFFSET: 0		MULTIPLE ACCT SELECT: N		(Y/N)			
USER:									
RECORD LAST CHANGED: 03/10/14 17:16 BY USER: 0255 , 00000255 ADD									
***** BASE24 *****									
NEW PAGE:		FILE DESTINATION:		NEW LOGICAL NETWORK ID:					
		F12-HELP							
RECORD RETRIEVED FROM \R2D2.\$TRAIN2.SYSFDATA.ATDD1								0000	

PTD Screen 1

PTD screen 1 contains the Dual Site Indicator field, as shown.

BASE24-POS		POS TERMINAL DATA		SYSF	EURO	03/10/16	11:33	01 OF 19
TERMINAL ID: 00000001				FIID: EURO				
				LOGICAL NET: SYSF				
RETAILER GROUP: EURO				REGION: POS				
LOCATION: MANUAL RETAIL ONLY				CITY: WATFORD				
STATE:		COUNTRY: GB		POSTAL CODE:				
TERMINAL TYPE: 41		(B24 STANDARD DEVICE)		PROTOCOL: X (X.25)				
TERMINAL PHONE: 00000000000000000000		BAUD RATE: 99 (N/A)						
TERMINAL OWNER: ACI WATFORD		TERM GROUP: EURO						
TIME OFFSET: 0		RELEASE #: 50						
TERM SIC CODE: 0		MAIL/PHONE SIC CODE: 0						
RETAILER ID: 21000001		RTTN: 00000000001						
TERM STATUS: A (ACTIVATED)		BILLING INFO:						
CLERK ID:		LANGUAGE ID: 0						
DH PROCESS NAME: P1A^SPDH				DFLT CHECK ID: (*****)				
ROUTING GROUP:				DUAL SITE INDICATOR: L (LOCAL)				
RECORD LAST CHANGED: 03/10/16 11:22				BY USER: 0255 , 00000255 CHANGE				
***** BASE24 *****								
NEW PAGE:		FILE DESTINATION:		NEW LOGICAL NETWORK ID:				
F12-HELP								
RECORD RETRIEVED FROM \R2D2.\$TRAIN2.SYSFDATA.PTDD1								0000

DCT

The ATM Device Control Terminal Server process has been enhanced to identify the location of particular terminal as local or remote and route requests accordingly.

Please see the BASE24 Release 6.0 Version 4 Implementation Guide for details of the LCONF params and assigns required by the DCT Server process for Dual Site Support.

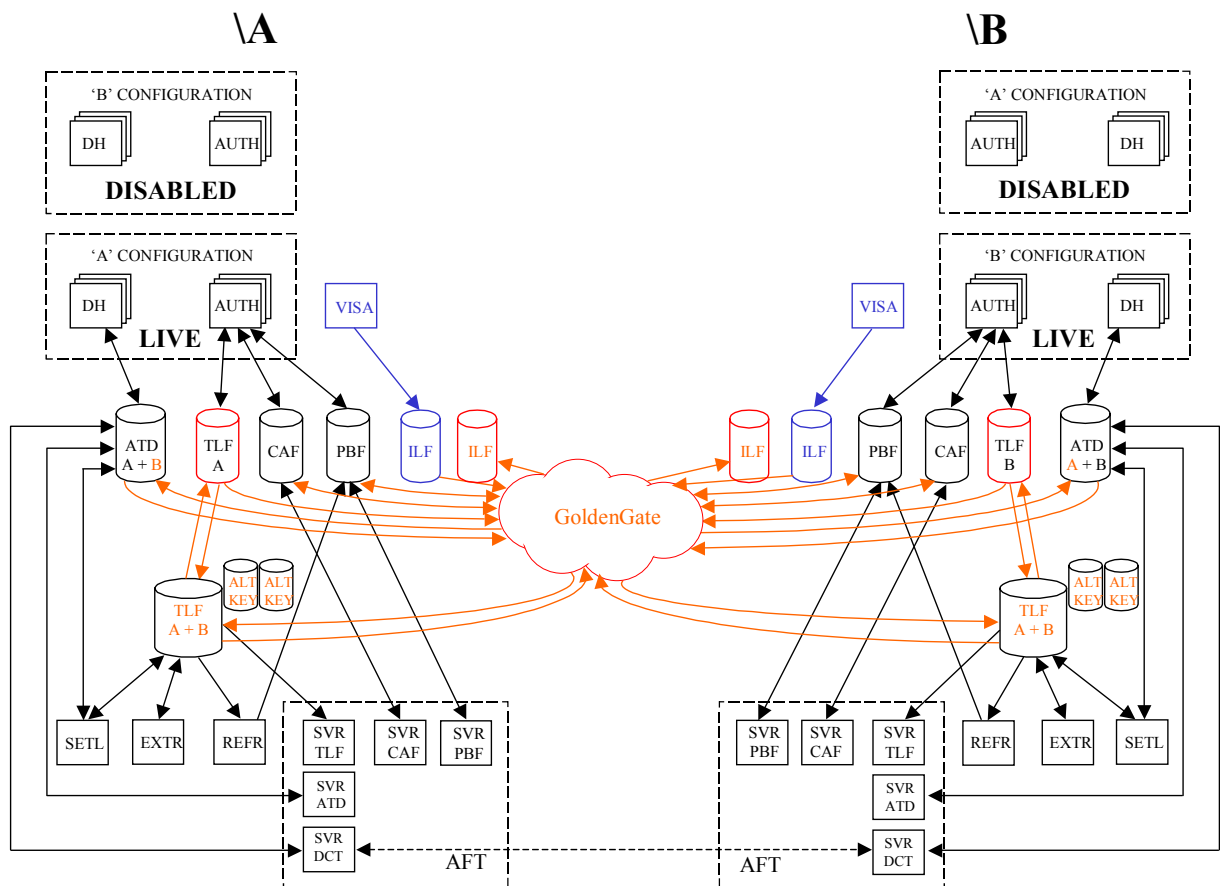
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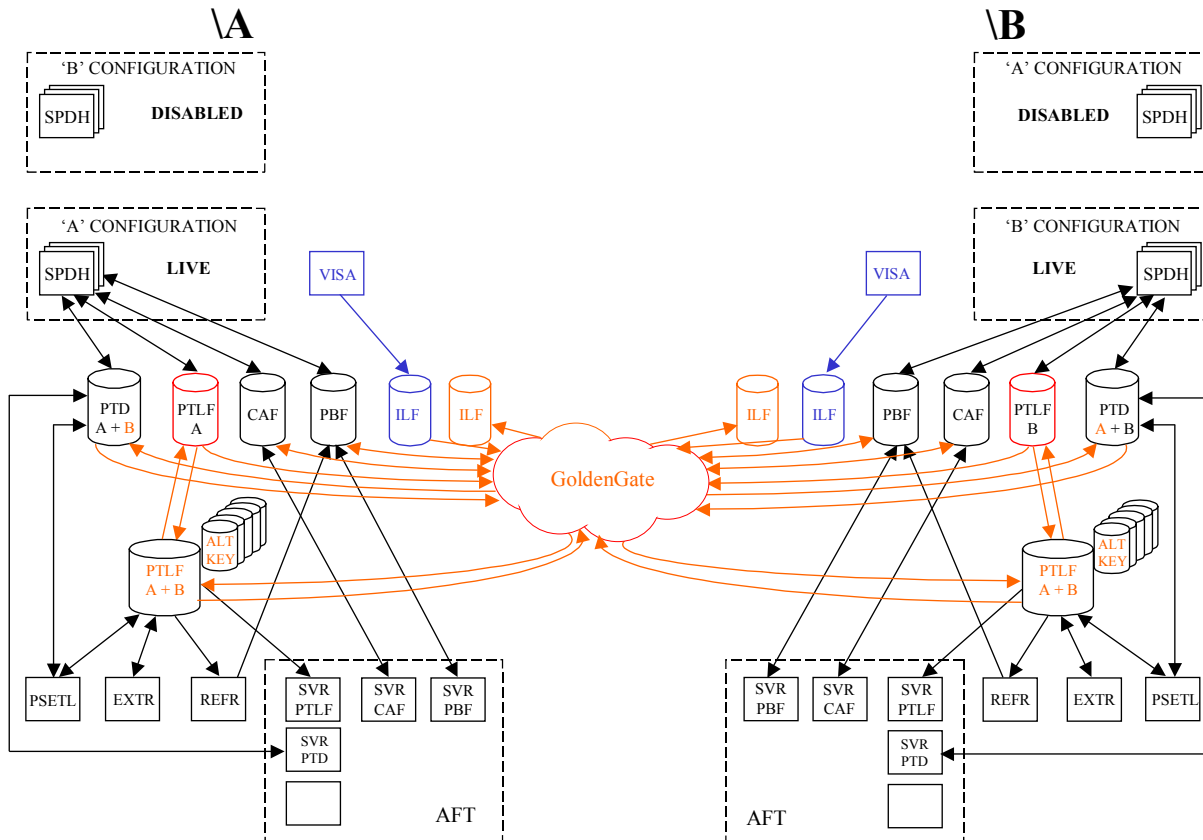
Dual Site Implementation Overview

This section discusses a variety of issues and requirements influencing Dual Site implementation and makes some recommendations as to a possible configuration approach.

The following diagram illustrates the files and processes involved when dual site processing is implemented for the BASE24-atm product.



The following diagram illustrates the files and processes involved when dual site processing is implemented for the BASE24-pos product.



Database Organization

Configuration files (LCONF, IDF, ICF, HCF, etc.) can easily be replicated by GoldenGate if the HP NonStop and BASE24 configurations on the two systems are identical. If there are differences in disc names or layouts on the two systems then GoldenGate can be used to replicate data and replace disc names as appropriate. For more significant configuration differences it may be required that these files be maintained independently on the two systems.

Security files (KEYF, xxx, xxx, xxx, etc.) can be replicated by GoldenGate if the HSM and BASE24 security configurations on the two systems are identical. Otherwise these files will need to be maintained independently on the two systems.

Single sets of authorization files (CAF, PBF, NEG, UAF) per Logical Network are required on each system. These files are synchronized by GoldenGate and do not require any special processing by BASE24. Each copy of BASE24 will access local copies referenced on the local LCONF and will be unaware of the remote copies or duplication undertaken by GoldenGate.

Transaction log files can be held as single copies per Logical Network on each system with new transaction records on each file being replicated to the alternate file by GoldenGate. This would result in BASE24 transaction records written by authorization processes being mixed with replicated records written by GoldenGate. This may be of concern for some organizations for audit or reconciliation reasons. Another approach would be to have 'prime' log files on each system only written to by BASE24 authorization processes and a merged log file on each system containing local and remote transaction records replicated by GoldenGate. With this option all settlement, extract, refresh, perusal and reporting activity could be supported by the merged log files. This would mean that the prime log files would not require alternate key files, as no perusal access would be required. This approach is recommended, and assumed throughout the rest of this document, although not required for Dual Site operation.

Interchange log files may or may not require replication depending upon the requirements for file access to support reconciliation. If required, then replication would typically be to and from single copies on either system, rather than bi-directional replication to and from ILFs containing local and remote records.

BASE24 Configuration

A BASE24 configuration needs to be built on each system. This will include BASE24 application configuration files such as the LCONF, IDF, ICF, HCF, initial copies of all BASE24 application data files and any template files required for future file creation. All files required for each system should be local to that system. One of the configurations may be the existing production system in which case care will need to be taken not to interrupt processing while building the second system. GoldenGate provides various tools to assist with this activity.

If the HP NonStop systems are duplicates and the BASE24 systems are to be duplicates then a configuration could be built on one system and replicated to the other via GoldenGate; if not, then separate configurations will need to be built and maintained. Special consideration should be given to security files.

Each system should have enough ATM Auth, POS Router / Auth and POS Device Handler / Router / Auth processes to handle the full transaction load if required. All such processes would probably be up and running at all times. Obviously this would result in them running at a low transaction rate and CPU utilization when both sites are operational, but would reduce the activities and time required to invoke single site operation in the event of an outage on the alternate site. If a smaller number of these processes are required to be run on each site during dual site operations, then when single site operation is invoked and additional such processes are started, different LCONF notify lists would be required. This could be resolved by the use of the new broadcast notify capability.

ATM Device Handler processes could be configured using one of two ways. The first approach would be to allocate the ATMs on each system to a particular set of Device Handlers, which would only be in operational use on one system during normal dual site running. Each Device Handler would also have a copy running on the other system, with the same ATMs configured, but this 'backup' copy would only be used in single site operations. During dual site operation the backup copies could be running (but with no started ATMs and no activity), stopped, or even disabled according to operational preferences and requirements. Having the backup processes running would reduce the activities and time taken to invoke single site operation.

The second approach for the configuration of ATM Device Handlers is to configure duplicate operational sets of Device Handlers on each system, with local and remote ATMs evenly distributed between them. In this scenario, all Device Handlers would always be running, but would only have approximately half of their configured ATMs started during dual site operations.

ATD / PTD Configuration

Terminals will need to be allocated to each of the two systems. Although this will often be approximately a 50/50 distribution, this is not required. Often terminals (particularly ATMs) in close proximity will be allocated across the two systems. For instance two ATMs in the same branch will often be allocated one to each system to minimize the impact of any system outage.

Having allocated the terminals to the sites the setting of the local / remote flag will need to be considered. This can be achieved by ensuring AFT access is to a particular system then setting the DUAL SITE INDICATOR on screen 1 of the ATD / PTD to "L" (LOCAL) or "R" (REMOTE) as appropriate. This could also be achieved via a custom program if required. If the initial connectivity of the terminals on the Dual Site systems reflects their intended configuration, then the system can automatically configure the local / remote flags appropriately. This is possible because " " (space) in the local / remote field is treated as local. As " " is the default value, all records are effectively set to local. If GoldenGate is configured to switch "L" or " " to "R" on replication, then all local records will retain the " " setting, which will work with all processes accessing the field, and all remote records will be set to "R".

GoldenGate should be configured to maintain the local / remote flag during the initial setting and any subsequent file activity. GoldenGate should always switch the flag when replicating changes to the alternate system. See the GoldenGate Dual Site Processing Guide for details of how to configure this.

Note that if one system becomes unavailable and terminals are to be temporarily run on the alternate system, there is no need to update the local / remote flag. The 'remote' terminals will be started and run on the local system and settlement will be configured to cutover all ("A") terminals.

TLF / PTLF Perusal

The merged TLF or PTLF on each system will contain copies of all transactions regardless of the system of origin. Therefore users can connect to either system and still have visibility to all transactions.

Remote transactions (transactions authorized on the alternate system and replicated to the local system by GoldenGate) can be identified on the TLF / PTLF detail display screens (screens 2 and 4 respectively).

This is achieved via use of the generic DUAL-SITE-DISPLAY LCONF param, which also requires the setting of process specific (SERVER-TLF and / or SERVER-PTLF) assigns for TLF or POS-PTLF.

If a merged TLF / PTLF is not being used, and remote transactions are replicated to the local prime TLF / PTLF by GoldenGate, then the generic DUAL-SITE-DISPLAY LCONF param can be used to request the identification of remote transactions on the TLF / PTLF detail display screens. In this case, the process specific TLF and / or POS-PTLF assigns will be required even though they will point to the same location as the generic assigns.

Settlement

Settlement can be configured to cutover local terminals only, remote terminals only or all terminals by setting the DUAL-SITE-MODE LCONF param to “L”, “R” or “A” respectively. In the example configuration described above, a setting of “L” on both systems is recommended. This will cause settlement to run on both systems, each copy cutting over local terminals on the ATD / PTD and writing records to the local merged (P)TLF. Remote terminal records on the local ATD / PTD will also be read, but will not be processed. Records written by Settlement will be replicated to the remote merged (P)TLF and the local prime (P)TLF by GoldenGate.

In the event of the failure of one system, the DUAL-SITE-MODE LCONF param for Settlement would be set to “A” on the surviving system to cutover all terminals.

Refresh

Refreshing of Authorization files (CAF, PBF, NEG, etc) can be undertaken on a single system, with changes replicated to the alternate system by GoldenGate, or undertaken on both systems. Factors influencing this decision will undoubtedly include the number of records being refreshed via full or partial refreshes, the capacity and utilization of the connection between the systems and the ability to deliver refresh files to the two systems in a timely manner.

If single site refreshing is to be used, then the DUAL-SITE-MODE LCONF param for Refresh should be set to “A”, to impact all transaction records, and all changes replicated to the alternate system via GoldenGate. If Dual Site refreshing is to be used, then the DUAL-SITE-MODE LCONF param for Refresh should be set to “L” on each site, to impact only local transaction records, and all impacting changes (but not refreshed records) should be replicated to the alternate system via GoldenGate. See the GoldenGate Dual Site Processing Guide for details of how to configure GoldenGate for these scenarios.

Extract

Extract of transaction log files (TLF, PTLF) can be undertaken on a single system or both systems. If a single system, then the system undertaking the extract could be fixed, or could switch between the two systems. This decision will primarily be driven by the requirements of the receiving host system.

If single site Extract is to be used, then the DUAL-SITE-MODE LCONF param for Extract should be set to “A”, to extract all transaction records, and changes to the extract record in the (P)TLF replicated to the alternate system via GoldenGate. If Dual Site extracting is to be used, then the DUAL-SITE-MODE LCONF param for Extract should be set to “L” on each site, to extract only local transaction records, and changes to the extract record in the (P)TLF should not be replicated to the alternate system via GoldenGate. See the GoldenGate Dual Site Processing Guide for details of how to configure GoldenGate for these scenarios.

DCT

DCT Dual Site support provides for correct communication with the appropriate Device Handler for the ATM in question. This is achieved by accessing the local ATD and determining whether the ATM is local or remote. If local, the local Device Handler is called as normal. If remote, the request is passed to the DCT on the remote system.

LCONF assigns REMOTE-PMON and REMOTE-LCONF and LCONF param DUAL-SITE-MODE will be required to configure this.

Reporting

Standard BASE24 reports have not been enhanced to support Dual Site processing. If standard reports are run on either system, all transactions will be processed. Therefore, if standard reports are used in a Dual Site environment, they should be run on one or other system and will include all transactions.

Multiple Time Zone Issues

Various issues will arise if the two dual site systems are operating in different time zones. In such cases, consideration should be given to running both systems under the same time zone and configuring time offsets to individual devices as appropriate. This would enable the synchronization of Logical Network cutover, Extract and Refresh timestamp processing to enable processing transactions for both dual sites.

Other solutions involving multiple Logical Networks may be necessary if true time zone differentiation is required.

Single Site Invocation

If single site operations are required for any reason, then it should be noted that there is no need to reconfigure ATD / PTD records to identify all terminals as local on the operational system. The Settlement, Refresh and Extract DUAL-SITE-MODE parameters will need to be set to "A", and processes warmbooted as appropriate, to ensure all terminals and transactions are correctly processed.

If additional ATM Device Handler, ATM Auth, POS Router / Auth or POS Device Handler / Router / Auth processes are required for single site operations (see BASE24 Configuration section above), then they will need to be started. If this is the case, and broadcast notification has not been used, then LCONF notify lists will need to be updated and processes warmbooted as appropriate.

GoldenGate configuration changes may also be required. See the GoldenGate Dual Site Processing Guide for details.

The major effort required to invoke single site operations will be the switching of all communications connections and terminals to the operational site. This will involve the starting of the appropriate lines and stations through SCF (or similar HP interfaces) and NCPCOM / NCS and the physical establishment of communications connections to the single site system.

During the time that this takes, production operations on the terminals and BASE24 application processes already in use on the single site system will be unaffected. Thus, at no point will a full system outage be experienced. The time taken to restore full operations will be minimized by well-documented and rehearsed procedures.

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Appendix A

LCONF Assigns and Params

The following sections identify the LCONF Assigns and Params associated with the BASE24 Release 6.0 Version 4 Dual Site support enhancements.

Assigns

POS-PTLF – Used by POS Settlement, Extract, Refresh and Server-PTLF if a PTLF other than that in the Generic LCONF POS-PTLF is to be used by these processes.

REMOTE-LCONF – Used by DCT to access remote Pathway in order to communicate with remote Device Handler managing remotely connected ATM.

REMOTE-PMON – Used by DCT to access remote Pathway in order to communicate with remote Device Handler managing remotely connected ATM

TLF – Used by ATM Settlement, Extract, Refresh and Server-TLF if a TLF other than that in the Generic LCONF TLF is to be used by these processes.

Params

BROADCAST-DELAY – Used by Settlement to control delay period after new (P)TLF creation prior to sending of notify messages to enable creation of GoldenGate copies of file to complete. Used by Refresh to control delay after sending of notify messages to completion of impacting.

BROADCAST-NOTIFY - -Used by Settlement and Refresh to control broadcast of notify messages.

DUAL-SITE-DISPLAY – Used by Server-TLF and Server-PTLF to control identification of remote records on detail display.

DUAL-SITE-MODE – Used by Settlement, Extract, Refresh and DCT to control access to and management of local and remote records.

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